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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE/SENIOR SERTIFIKAAT
NATIONAL SENIOR CERTIFICATE/
NASIONALE SENIOR SERTIFIKAAT**

GRADE/GRAAD 12

**MATHEMATICAL LITERACY P1/
WISKUNDIGE GELETTERDHEID V1**

NOVEMBER 2020

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
M	Method/Metode
MA	Method with accuracy/Metode met akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
A	Accuracy/Akkuraatheid
C	Conversion/Herleiding
S	Simplification/Vereenvoudiging
RT	Reading from a table/graph/document/diagram/Lees vanaf tabel/grafiek/dokument/diagram
SF	Correct substitution in a formula/Korrekte vervanging in 'n formule
O	Opinion/Explanation/Opinie/Verduideliking
P	Penalty, e.g. for no units, incorrect rounding off, etc./Penalisasie, bv. vir geen eenhede, verkeerde afronding, ens.
R	Rounding off/Afronding
NPR	No penalty for rounding/Geen penalisasie vir afronding nie
AO	Answer only/Slegs antwoord
MCA	Method with consistent accuracy/Metode met volgehoue akkuraatheid
RCA	Rounding consistent with accuracy/ Afronding met volgehoue akkuraatheid

**This marking guideline consists of 17 pages.
Hierdie nasienriglyne bestaan uit 17 bladsyes.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- CA marks only apply if at least 1 correct value is used.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die tweede berekeningsfout.
- CA geld alleenlik wanneer ten minste 1 korrekte waarde gebruik is.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra item.

QUESTION/VRAAG 1 [30 MARKS/PUNTE] ANSWER ONLY FULL MARKS			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
1.1.1	Vertical bar graph/Vertikale staafgrafiek. Bar/Balk/Staaf, Column graph/Kolomgrafiek ✓✓A	2A bar graph (2)	D L1
1.1.2	✓MA $A = R110 + R11$ $= R121$ ✓CA	1MA adding correct values 1CA Simplification (2)	F L1
1.1.3	✓MA $B = R141 - R126$ $= R15$ ✓CA	1MA subtracting correct values 1CA simplification (2)	F L1
1.1.4	Difference/Verskil $R126 - R110$ ✓MA $= R16$ ✓A	1MA subtract lowest from highest 1A simplification (2)	F L1
1.1.5	Increased Delivery fee/Verhoogde afleweringsooi $= R10,00 \times 6,32\%$ ✓MA $= R0,632$ $= R0,63$ ✓A OR/OF $= R10,00 \times \frac{6,32}{100}$ ✓M $= R0,632$ $= R0,63$ ✓A OR/OF	1MA calculating percentage 1A simplification OR/OF	F L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
1.1.5	Increased delivery fee/<i>Verhoogde afleveringskoste</i> $= R10 \times 1,0632$ ✓MA $= R10,632$ Increase in delivery fee/<i>Verhooging in afleveringskoste</i> $= R10,63 - R10,00$ $= R0,63$ ✓A	1MA calculating percentage 1A simplification (2)	
1.2.1	2008 ✓✓RT	2RT reading correct year (2)	D L1
1.2.2	✓MA ✓RT Difference/<i>Verskil</i> $= R11,04 - R4,31$ $= R6,73$ ✓CA	1MA subtracting correct values 1RT correct values 1CA simplification (3)	F L1
1.2.3	✓MA $5,56 : 12,48$ ✓RT $1 : 2,24$ OR/OF $0,45 : 1$ ✓CA	1MA concept of ratio in correct order 1RT correct values 1CA simplification (3)	F L1
1.2.4	Total/<i>Totaal</i> $= 13,45 \times R4,00$ ✓MA $= R53,80$ ✓CA OR/OF $R : \ell$ $4 : 1$ ✓MA $53,80 : 13,45$ Total cost = R53,80 ✓CA	1MA multiplying correct values 1CA simplification (2)	F L1
1.2.5	2007 ✓✓RT	2RT reading correct year (2)	D L1

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
1.3.1	✓A ✓A Strip Map (Chart)/Strookkaart ✓✓A	2A strip map (chart) (2)	MP L1
1.3.2	Distance in metre/Afstand in meter = $779 \times 1\,000$ ✓MA = 779 000 ✓A	1MA multiplying by 1 000 1A simplifying NPU (2)	M L1
1.3.3 (a)	✓A ✓A Ladismith AND/EN Calitzdorp	1A correct town 1A correct town (2)	MP L1
1.3.3 (b)	The distance from Riversdale to Oudtshoorn/ <i>Afstand vanaf Riversdal na Oudtshoorn</i> = 82 km + 45 km + 53 km ✓MA = 180 km ✓CA	1MA adding correct values 1CA simplification (2)	MP L1
		[30]	

QUESTION/VRAAG 2 [42 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.1.1	R4 656,71 ✓✓A	2A correct balance (2)	F L1
2.1.2	Full date/Volle datum 1 February/Februarie 2019 ✓✓A 01/02/19 01/02/2019	2A full date (2)	F L1
2.1.3	R1 215,36 ✓✓A	2A correct amount (2)	F L1
2.1.4	R3 750,00 ✓✓A	2A correct amount (2)	F L1
2.1.5	FNB electronic payments/ENB elektroniese betaling ✓RT ✓RT R101,99 + R698,01 = R800,00 ✓A	1RT 1 st value correct 1RT 2 nd value correct 1A simplification AO (3)	F L1
2.1.6	Price excluding VAT/Prys BTW uitgesluit ✓RT = R4000,00 × $\frac{100}{115}$ ✓MA = R3 478,26 ✓CA OR/OF Price excluding VAT/Prys BTW uitgesluit ✓RT $\frac{R4000}{1,15}$ ✓MA = R3 478,26 ✓CA OR/OF $\text{VAT amount/BTW bedrag} = R4000,00 \times \frac{15}{115}$ ✓RT ✓MA = R521,74 Price excluding VAT/Prys BTW uitgesluit = R4 000 – R521,74 = R3 478,26 ✓CA	1RT price of item 1MA calculating VAT 1CA price excluding VAT AO (3)	F L2

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.2.1	South African Revenue Services/SARS Revenue Services ✓✓A <i>Suid Afrikaanse Inkomstedienste/SAID</i> <i>Inkomste(belasting)dienste</i>	2A name (2)	F L1
2.2.2	2 / TWO / TWEE ✓✓A OR/OF 7 / SEVEN / SEWE	2A correct bracket (2)	F L1
2.2.3	Annual tax before rebates/ <i>Jaarlikse inkomstebelasting voor belastingkortings</i> = R35 253 + 26% of taxable income above 195 850 = R35 253 + 26% × (R305 174,44 – R195 850) ✓SF = R35 253 + R28 424,35 ✓M = R63 677,35 ✓CA Monthly tax before rebates/ <i>Maandelikse inkomstebelasting voor belastingkortings</i> = R63 677,35 ÷ 12 ✓MCA = R5 306,45 ✓CA OR/OF Annual tax before rebates/ <i>Jaarlikse inkomstebelasting voor belastingkortings</i> = R532 041 + 45% of taxable income above 1 500 000 = R532 041 + 45% × (R3 662 093,28 – R1 500 000) ✓SF = R532 041 + R972 941,98 ✓M = R1 504 982,98 ✓CA Monthly tax before rebates/ <i>Maandelikse inkomstebelasting voor belastingkortings</i> = R1 504 982,98 ÷ 12 ✓MCA = R125 415,25 ✓CA	CA from question 2.2.2 1SF correct substitution 1M adding correct amounts 1CA simplification 1MCA dividing by 12 1CA simplification NPR (5)	F L3
2.2.4(a)	✓✓RT Primary rebate/ <i>Primêre korting</i> OR/OF R14 067,00	2RT reading from the table (2)	F L1
2.2.4(b)	3/THREE/DRIE ✓✓A	2A correct number of rebates (2)	F L1

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
2.3.1	Selling price of one photo/<i>Verkoopprys van een foto</i> $\checkmark$MA $\frac{R500}{25}$ OR $\frac{R1000}{50}$ OR $\frac{R1600}{80}$ OR $\frac{R2500}{125}$ OR $\frac{R3000}{150}$ $= R20 \checkmark$A OR/OF $R4\,000 \div 200 \checkmark$MA $= R20 \checkmark$A	1MA dividing 1A simplification AO (2)	F L1
2.3.2	Total income received/<i>Totale inkomste ontvang:</i> $\checkmark$CA $\checkmark$A Income = $R20,00 \times n$, where n = number of photos Income = $R20,00 \times$ number of photos <i>Inkomste = $R20,00 \times n$, waar n = aantal foto's</i> <i>Inkomste = $R20,00 \times$ aantal foto's</i>	CA from Question 2.3.1 1CA $R20,00$ 1A multiply by unknown (2)	F L2
2.3.3 (a)	$R5,00 \checkmark\checkmark$A	2A variable cost NPU (2)	F L1
2.3.3 (b)	A : Expenses = $R1\,125 +$ number of photos $\times R5,00$ A : <i>Uitgawes = $R1\,125 +$ aantal foto's $\times R5,00$</i> $\checkmark$SF $A = R1\,125 + (80 \times R5,00)$ $A = R1\,125 + R400 \checkmark$MCA $= R1\,525 \checkmark$CA	1SF substituting value 1MCA adding values 1CA simplification AO (3)	F L2
2.3.4 (a)	$\checkmark\checkmark$A Income and expenses of Ella's photography business <i>Inkomste en uitgawes van Ella se fotografiebesigheid</i>	2A correct heading (2)	F L1
2.3.4 (b)	X $\checkmark\checkmark$A	2A correct graph (2)	F L1
2.3.4 (c)	75 photographs/<i>foto's</i> $\checkmark\checkmark$A	2A correct number of photographs (2)	F L1
		[42]	

QUESTION/VRAAG 3 [31 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
3.1.1	<u>Legs of ottomans/Pote van ottomans:</u> 2 cubic/kubieke ottomans $\times$ 4 legs/pote = 8 legs/pote ✓A 1 retangular/reghoekige ottoman $\times$ 6 legs/pote = 6 legs/pote 8 + 6 ✓MA = 14 legs/pote ✓CA	1A number of legs 1MA adding 6 legs 1CA total number of legs AO (3)	M L1
3.1.2	$\text{Radius} = \frac{75 \text{ mm}}{2} \quad \checkmark \text{MA}$ $= 37,5 \text{ mm} / 3,75 \text{ cm} \quad \checkmark \text{A}$	1MA concept of radius 1A simplification AO NPR (2)	M L1
3.1.3	Total height/Totale hoogte: 50 cm + 12 cm ✓C = 62cm ✓A OR/OF Total height/Totale hoogte: = 120 mm + 500 mm = 620 mm ✓A = 62 cm ✓C	1C converting to cm 1A finding the height AO (2)	M L1

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
M L2	Area/Oppervlakte ✓A ✓A $(50\text{cm} \times 50\text{cm}) + (120\text{cm} \times 50\text{cm})$ $2\,500\text{ cm}^2 + 6\,000\text{ cm}^2$ Total Area/Totale Oppervlakte $(10 \times 2\,500\text{ cm}^2) + (2 \times 6\,000\text{ cm}^2)$ ✓M $25\,000\text{ cm}^2 + 12\,000\text{ cm}^2$ ✓M $37\,000\text{ cm}^2$ ✓CA OR/OF 8 square sides/vierkantige sye $\times (50 \times 50)$ $= 20\,000\text{ cm}^2$ ✓A 2 rectangular sides/reghoekige sye $\times (120 \times 50)$ $= 12\,000\text{ cm}^2$ ✓A 2 square sides / vierkantige sye $\times (50 \times 50)$ $= 5\,000\text{ cm}^2$ ✓A Total area to be painted/Totale area wat geverf moet word: $= 20\,000\text{ cm}^2 + 12\,000\text{ cm}^2 + 5\,000\text{ cm}^2$ ✓M $= 37\,000\text{ cm}^2$ ✓MA OR/OF Total perimeter/Totale Omtrek ✓A ✓M $= (50+50+50+50+50+50+50+50+120+50+50+120)\text{ cm}$ $= 740\text{ cm}$ ✓A Total area to be painted/Totale area wat geverf moet word: $= 740\text{ cm} \times 50\text{ cm}$ ✓MA $= 37\,000\text{ cm}^2$ ✓A	1A area 1A area 1M multiplying correct values 1M adding the two areas 1CA simplification OR/OF 1A simplification 1A simplification 1A simplification 1M adding all values 1MA finding total area OR/OF 1A all correct values 1M adding correct values 1A simplification 1MA multiplying correct values 1A simplification (5)	

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.1.5	$37\,000\text{ cm}^2 \div 10\,000 = 3,7\text{ m}^2 \checkmark\text{C}$ Total area to be painted/ <i>Totale area wat geverf moet word</i> $= 3,7\text{ m}^2 \times 2 \checkmark\text{M}$ $= 7,4\text{ m}^2$ Spread rate/ <i>sprydingskoers</i> $\frac{7,4\text{ m}^2}{8\text{ m}^2} \times 1\,000 \checkmark\text{M}$ $= 925\text{ millilitres/milliliter} \checkmark\text{CA}$ OR/OF Spread rate/ <i>sprydingskoers</i> $= 8 \times 10\,000\text{ cm}^2/\ell$ $= 80\,000\text{ cm}^2/\ell \checkmark\text{M}$ Amount of paint / <i>aantal verf in ℓ</i> $= \frac{37\,000}{80\,000} \checkmark\text{M}$ $= 0,4625$ Amount of paint for 1 coat / <i>aantal verf vir 1 deklaag in mℓ</i> $= 0,4625 \times 1\,000$ $= 462,5 \checkmark\text{C}$ Amount of paint for 2 coats/ <i>aantal verf vir twee deklae</i> $= 462,5\text{ mℓ} \times 2$ $= 925\text{ mℓ} \checkmark\text{CA}$ OR/OF Total area to be painted/ <i>Totale area wat geverf moet word:</i> $= 37\,000\text{ cm}^2 \div (100)^2 = 3,7\text{ m}^2 \checkmark\text{C}$ Amount of paint for 1 coat/ <i>aantal ver vir 1 deklaag in ℓ</i> $= \frac{3,7}{8} \times 1 \checkmark\text{M}$ $= 0,4625\text{ ℓ}$ Total amount of paint/ <i>Totale aantal verfl</i> $= 0,4625 \times 1000 \times 2 \checkmark\text{M}$ $= 925\text{ mℓ} \checkmark\text{CA}$ OR/OF	CA Question 3.1.4 1C converting from cm^2 to m^2 1M area for 2 coats 1M divide by spread rate 1CA answer in millilitres OR/OF 1M multiplying by 8 1M dividing by 80 000 1C converting 1CA simplification OR/OF 1C conversion 1M dividing by 8 1M area of 2 coats 1CA simplification OR/OF	M L2

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.2.2	$P(\text{not selecting red material}) = \frac{6}{9} \checkmark A$ $= \frac{2}{3} \checkmark CA$ OR/OF $P(\text{not selecting red material}) = 1 - \frac{3}{9}$ $= \frac{6}{9} \checkmark A$ $= \frac{2}{3} \checkmark CA$	1A numerator 1A denominator 1CA simplification (3)	P L2
3.3.1	$1 \text{ inch} = 153,6 \div 60 \checkmark M$ $= 2,56 \text{ cm} \checkmark A$ OR/OF Alternative solution method: $\begin{array}{lcl} \text{inch} & : & \text{cm} \\ 60 & : & 153,6 \checkmark M \\ 1 & : & 2,56 \checkmark A \end{array}$ $1 \text{ inch} = 2,56 \text{ cm}$	1M dividing by 60 1A simplification (2)	M L1
3.3.2	$\text{Perimeter/Omtrek} = 2 \times (5 \text{ m} + 153,6 \text{ cm}) \checkmark RT$ $= 2 \times (500 \text{ cm} + 153,6 \text{ cm}) \checkmark C$ $= 1\,307,2 \text{ cm} \checkmark CA$ OR/OF $\text{Perimeter/Omtrek} = 5 \text{ m} + 5 \text{ m} + 153,6 \text{ cm} + 153,6 \text{ cm} \checkmark RT$ $= (500 + 500 + 153,6 + 153,6) \text{ cm} \checkmark C$ $= 1\,307,2 \text{ cm} \checkmark CA$	1RT correct value – 153,6 cm 1C converting from 5 m to cm 1CA simplification OR/OF 1RT correct value – 153,6 cm 1C converting from 5 m to cm 1CA simplification (3)	M L2
		[31]	

QUESTION/VRAAG 4 [17 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.1.1	R46 ✓✓A	2A name of route (2)	MP L1
4.1.2	Number scale OR Numeric scale OR Ratio scale ✓✓ A <i>Nommerskaal OF verhoudingskaal OF Getalskaal OF</i> <i>Numeriese OF Getalle Skaal OF Syferskaal</i>	2A identifying the scale (2)	MP L1
4.1.3	South West OR SW OR West of South West OR WSW ✓✓ A <i>Suidwes OF SW OF Wes van Suidwes OF WSW</i>	2A general direction (2)	MP L1
4.1.4	A = 210 km – (62 km + 13 km + 82 km) ✓MA A = 53 km ✓CA	1MA subtracting correct values 1CA simplification (2)	MP L1
4.1.5	Ladismith ✓✓A	2A correct town (2)	MP L2
4.2.1	Total length /Totale lengte ✓MA ✓MA = 20 cm + 229 cm + 20 cm + 20 cm + 229 cm + 20 cm = 538 cm ✓CA OR/OF Total length /Totale lengte ✓MA ✓MA 2 (20 cm + 229 cm + 20 cm) 2 × 269 cm = 538 cm ✓CA OR/OF Total length/Totale lengte ✓MA ✓MA = (20 cm × 4) + (229 cm × 2) = 80 cm + 458 cm = 538 cm ✓CA	1MA correct values (4×20) 1MA adding values (2×229) 1CA simplification OR/OF 1MA correct values (4×20) 1MA adding values (2×229) 1CA simplification OR / OF 1MA correct values (4×20) 1MA adding values (2×229) 1CA simplification (3)	MP L2

QUESTION/VRAAG 5 [30 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.1.1	TGA – team/span ✓✓RT	2RT correct tea (2)	D L1
5.1.2	Range/Omvang = 9,625 – 9,100 ✓RT = 0,525 ✓CA	1RT reading correct values 1CA concept of range (2)	D L1
5.1.3	Mean/Gemiddeld ✓RT = $\frac{9,100 + 9,250 + 9,300 + 8,650 + 9,100 + 9,050 + 8,750 + 9,050 + 8,300 + 9,200}{10}$ ✓M = 8,975 ✓CA	1RT correct values 1M concept of mean 1CA simplification NPR (3)	D L2
5.1.4	✓RT $A = 36,425 - (9,300 + 9,100 + 9,225)$ ✓M = 8,800 ✓A	1RT correct values 1M adding and subtracting 1A simplification (3)	D L1
5.1.5	36,425 ✓✓A	2A correct mode (2)	D L1
5.1.6	✓A $\frac{3}{5} \times 100\%$ ✓A = 60% ✓CA	1A numerator 1A denominator 1CA percentage NPR (3)	P L2
5.1.7	Quartile / Kwartiel 2 ✓RT = $\frac{9,375 + 9,400}{2}$ ✓M = 9,3875 ✓A	1RT arranging or correct values 1M dividing by 2 1A simplification NPR (3)	D L2

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.2.1	Fifty two million nine hundred and eighty two thousand. ✓✓A <i>Twee en vyftig miljoen negehonderd twee en tagtig duisend.</i>	2A amount in words (2)	D L1
5.2.2	Increase in population/ <i>Toename in bevolking</i> (2015-2016) ✓RT ✓M 56 020 718 – 54 901 943 = 1 118 775 ≈ 1 120 000 ✓R	1RT correct values 1M subtracting 1R correct rounding (3)	D L1
5.2.3	Annual population growth/ <i>Jaarlikse bevolkingstoename</i> (2015) $= \frac{54\,901\,943 - 53\,947\,998}{53\,947\,998} \times 100\%$ ✓SF = 1,768% ≈ 1,8% ✓CA	1SF substituting 54 901 943 1SF substituting 53 947 998 1CA simplification NPR (3)	D L2

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L																					
5.2.4	Estimated total population and annual growth from 2013-2017 <table><thead><tr><th>Year</th><th>Population (Number of people)</th><th>Population growth (in percentage)</th></tr></thead><tbody><tr><td>2013</td><td>52 982 000</td><td>~2.2</td></tr><tr><td>2014</td><td>53 947 998</td><td>~1.9</td></tr><tr><td>2015</td><td>54 901 943</td><td>~1.8</td></tr><tr><td>2016</td><td>56 020 718</td><td>~2.1</td></tr><tr><td>2017</td><td>56 521 948</td><td>~1.0</td></tr><tr><td>2018</td><td>57 725 606</td><td>~2.2</td></tr></tbody></table>		Year	Population (Number of people)	Population growth (in percentage)	2013	52 982 000	~2.2	2014	53 947 998	~1.9	2015	54 901 943	~1.8	2016	56 020 718	~2.1	2017	56 521 948	~1.0	2018	57 725 606	~2.2	D L2
Year	Population (Number of people)	Population growth (in percentage)																						
2013	52 982 000	~2.2																						
2014	53 947 998	~1.9																						
2015	54 901 943	~1.8																						
2016	56 020 718	~2.1																						
2017	56 521 948	~1.0																						
2018	57 725 606	~2.2																						
	1A – correctly plotted number of people 1CA – drawing of graph 1A – correctly plotted population growth 1CA – drawing of graph (4)																							
	[30]																							
	TOTAL/TOTAAL: 150																							